Reliance-Awareness Design and Evaluation in Explainable User Interfaces

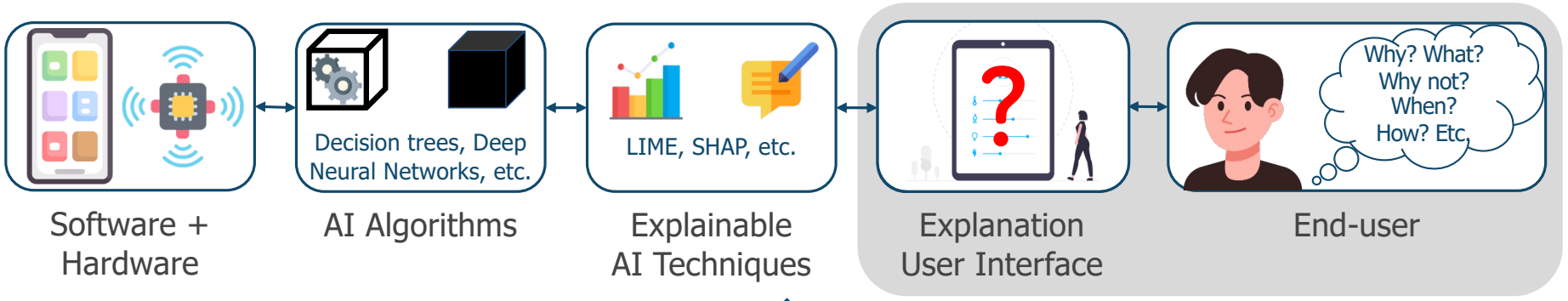
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IHM '25 - Atelier XAI4U
XAI4U – Explicabilité de l'IA pour les utilisateurs finaux

Research Context

Explainable AI (XAI) system components based on Gunning et al. [1]



Appropriate reliance: In decision-making processes end-users should know **when to rely** on AI's advice and **when not to rely** on it, avoiding over- and under-reliance [2]

Reliance-awareness – Our perspective: The capacity of **user interface design** to support the appropriate reliance of users in detecting correct recommendations and errors in XAI advice, ultimately supporting **informed decision-making**

[1] Gunning, D., Vorm, E., Wang, Y., & Turek, M. (2021). **DARPA's explainable AI (XAI) program: A retrospective**. Authorea Preprints.
[2] Naiseh, M., Al-Mansoori, R. S., Al-Thani, D., Jiang, N., & Ali, R. (2021). **Nudging through friction: An approach for calibrating trust in explainable ai**. In 8th International Conference on Behavioral and Social Computing (BESC) (pp. 1-5). IEEE.

Research Question

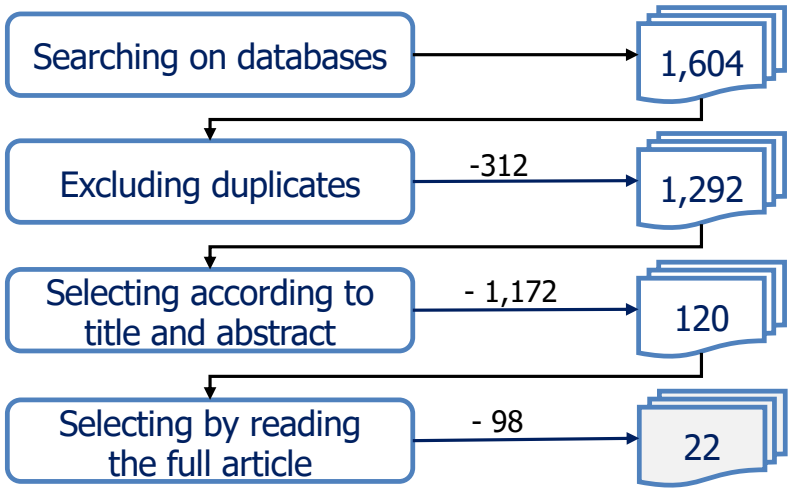
- How to design and evaluate explainable user interfaces to stimulate appropriate reliance in decision-making processes?

Goal

- To define an approach for reliance-awareness design and evaluation of explainable user interfaces through:
  - An evidence-based conceptual model
  - A design methodology

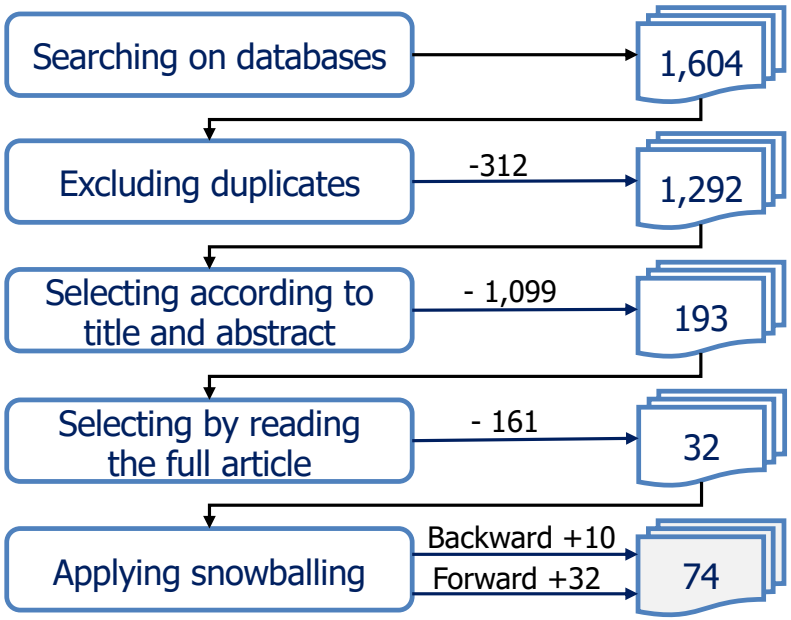
State-of-the-Art

Tertiary Study



There is no study on how to design and evaluate interactive explanation interfaces aware of reliance issues

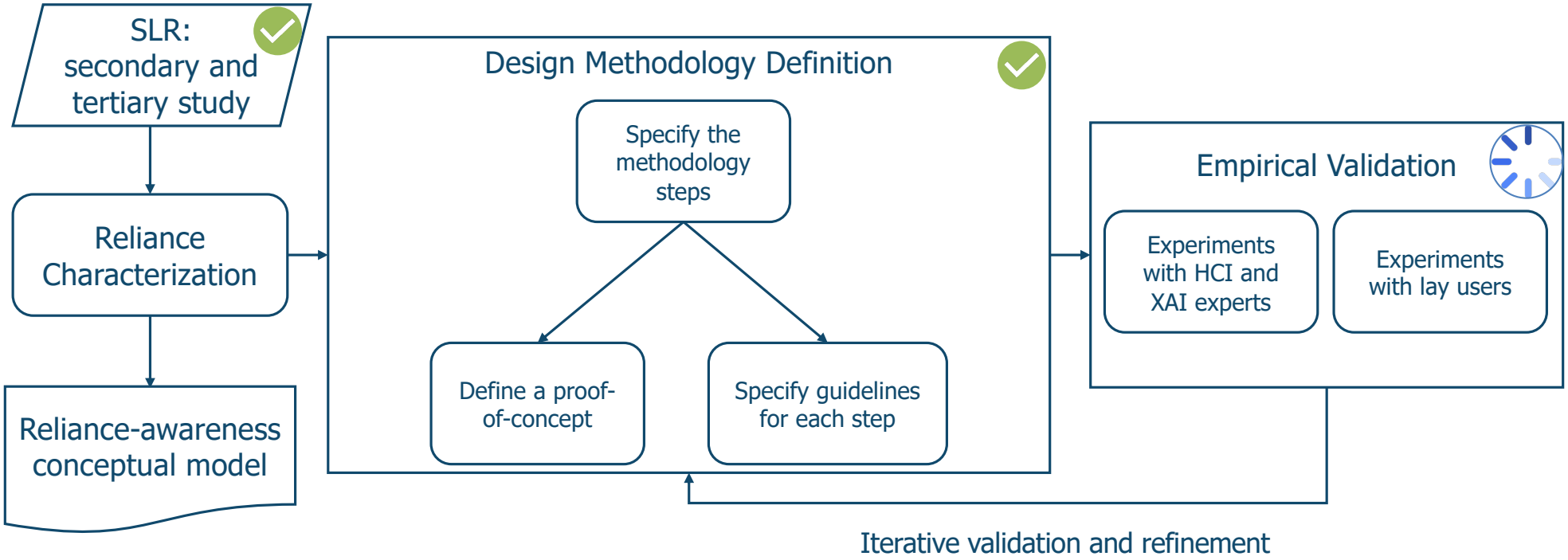
Secondary Study



de Souza Filho, J. C., Belloum, R., Oliveira, K. M. (2024). **Where Are We and Where Can We Go on the Road to Reliance-Aware Explainable User Interfaces?** In Proceedings of the 2024 IEEE Symposium on Visual Languages and Human-Centered Computing.

de Souza Filho, J. C., Belloum, R., Oliveira, K. M. (2025). **Characterizing appropriate reliance in XAI-assisted decision-making: Definitions, impact factors, and explanation design**. International Journal of Human-Computer Studies. Under Minor Revisions.

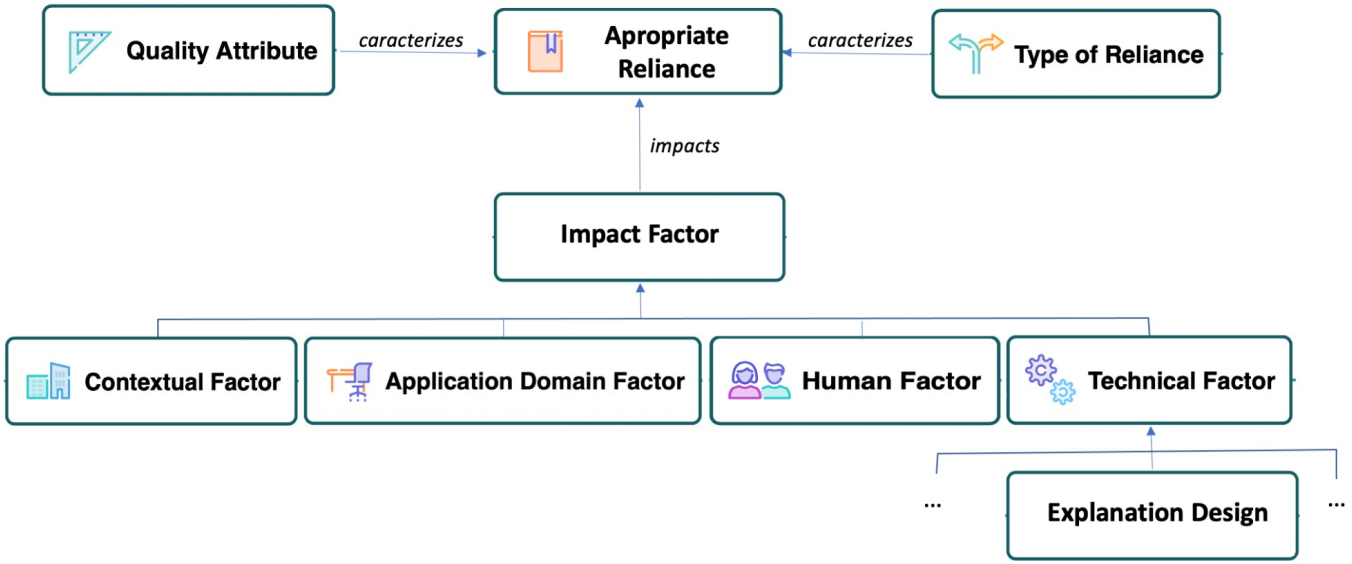
Methodology



Reliance-Awareness Conceptual Model

Insights from explanations designs:

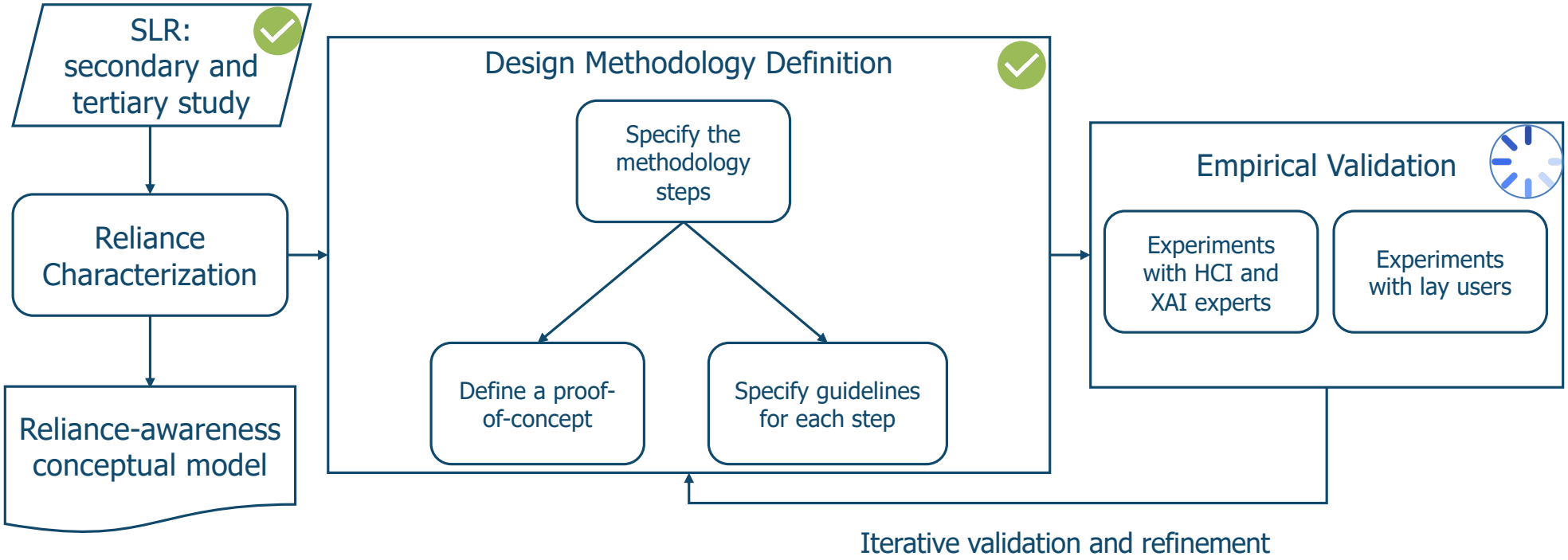
- ❑ Static (86.9%) and text-based (73.8%) explanations are dominant
- ❑ Business, education, and healthcare are well studied application domains
- ❑ LIME combined with textual and visual outputs in recommendation tasks are frequently linked to fostering appropriate reliance



de Souza Filho, J. C., Belloum, R., Oliveira, K. M. (2025). **How Does Explanation Design Influence Reliance Behaviors in XAI-Assisted Decision-Making?** Under-review for the 8th AAAI/ACM Conference on AI, Ethics, and Society (AIES 2025).

de Souza Filho, J. C., Belloum, R., Oliveira, K. M. (2025). **Characterizing appropriate reliance in XAI-assisted decision-making: Definitions, impact factors, and explanation design**. International Journal of Human-Computer Studies. Under Minor Revisions.

Methodology



RADIX:

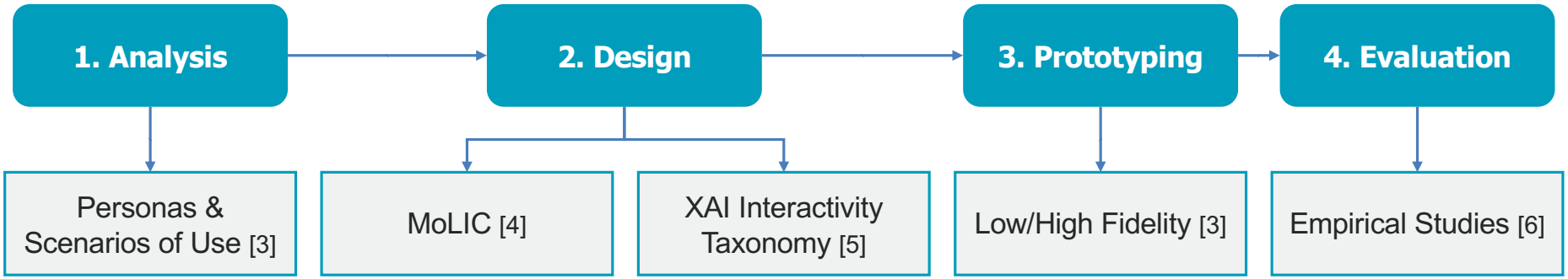
Reliance-Aware Design methodology for Interactive explanations in XAI systems



[3] Helen, S., Jenny, P., & Yvonne, R. (2019). **Interaction design: beyond human-computer interaction**.

RADIX:

Reliance-Aware Design methodology for Interactive explanations in XAI systems



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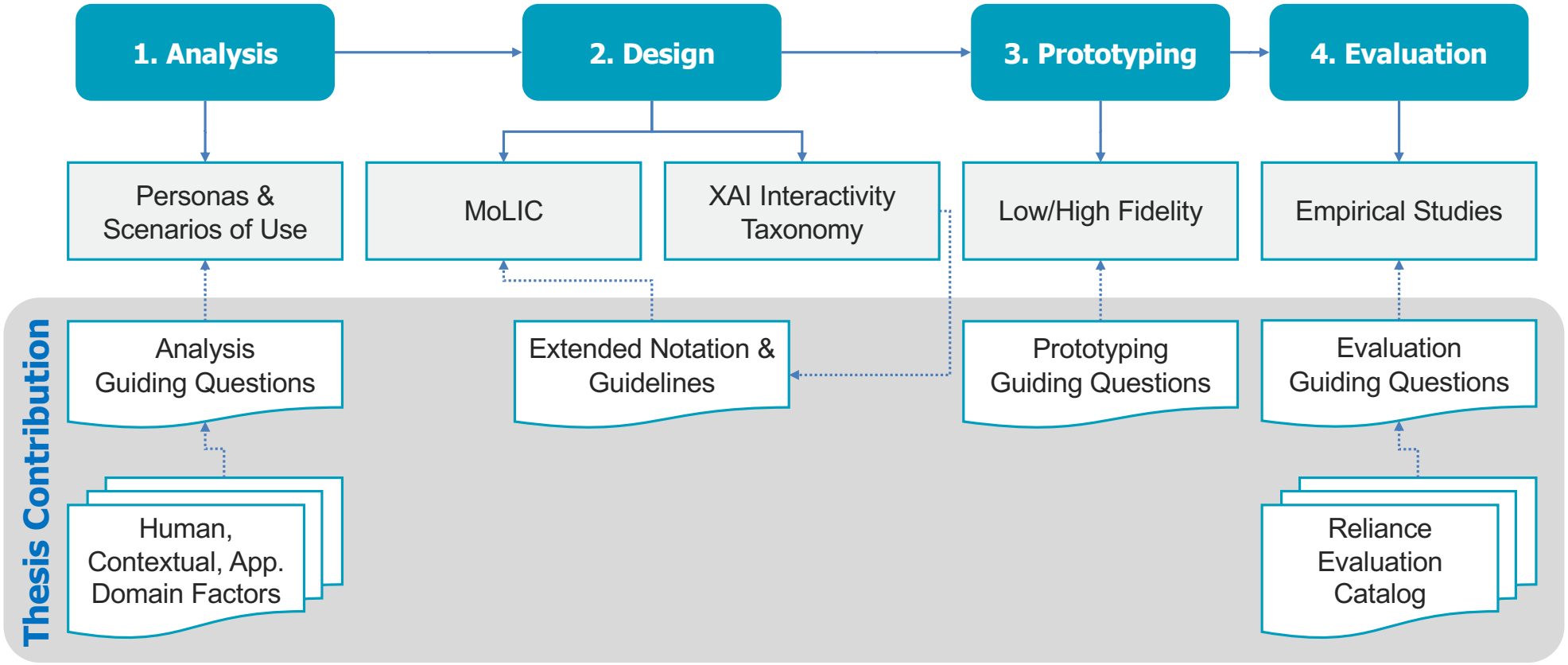
[4] Guimarães, C. L. (2023). **Realigning MoLIC to the Interaction-as-Conversation Metaphor**.

[5] Bertrand, A., Viard, T., Belloum, R., Eagan, J. R., & Maxwell, W. (2023). **On selective, mutable and dialogic XAI: A review of what users say about different types of interactive explanations**. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems.

[6] Rong, Y., Leemann, T., Nguyen, T. T., Fiedler, L., Qian, P., & Unhelkar, V. (2023). **Towards human-centered explainable ai: A survey of user studies for model explanations**. IEEE transactions on pattern analysis and machine intelligence, 46(4), 2104-2122.

RADIX:

Reliance-Aware Design methodology for Interactive explanations in XAI systems



Proof-of-Concept

Personas & Scenarios of Use

Persona



Regina has 35 years old. She works as a human resource recruiter, married, with one child. Her workplace is a big health center and every week she has to analyze several candidates' biographies for hiring process. Given the amount of process and candidates, she has to guess the candidate's profession quickly only considering their short biography.

Domain expertise: Expert.

Task familiarity: She carry out the biography prediction task for 5 years.

AI literacy: Lay user.

AI familiarity: It is her first experience using an AI system.

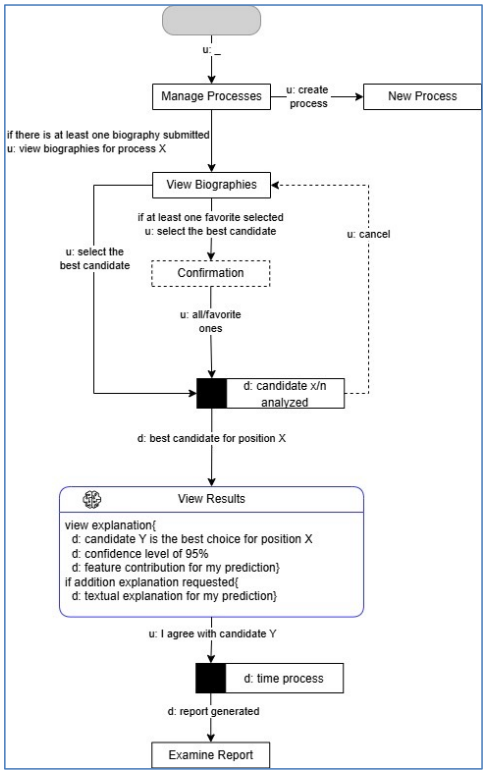
Propensity: She is open to try a new experience, however, she is skeptic about using an AI system to support her in a task she has been doing for many years.

Abilities: Since she is experience with the task and domain, she has a high ability to identify and correct system mistakes.

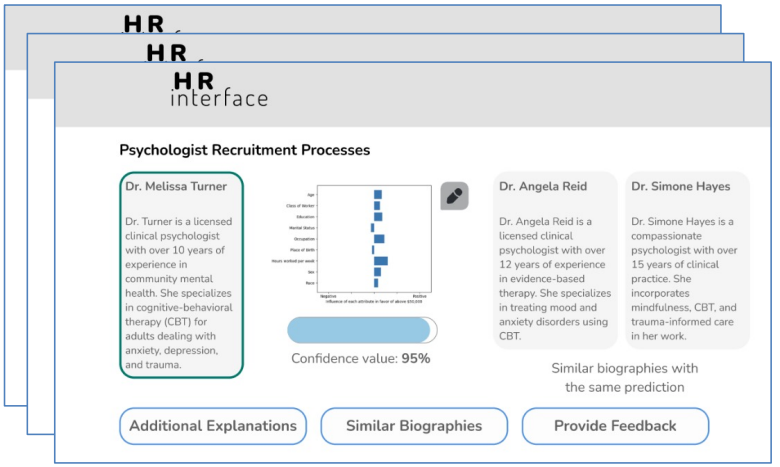
Scenario of Use

Regina has to analyze 20 short biography descriptions to quickly predict the candidate's profession for each biography. It stars when she authenticates into the HR system. Once authenticated, the system shows to Regina one biography per time and asks her to make an initial prediction according to 5 pre-defined healthcare professions, in addition to one open-ended field "Other", in which she can type another profession. Regina has to make this decision in 3 minutes for each biography. Then, for each biography per time, the system shows its prediction along with a confidence value, feature-based, and example-based explanations. Based on that, Regina has more 3 minutes to make a final decision for each biography prediction.

MoLIC



High Fidelity Prototype



Reliance-Awareness Design and Evaluation in Explainable User Interfaces

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Merci Beaucoup ☺